

UED Smart Display Controller

Assembly & Operation Guide

🔧 Hardware Overview & Assembly

This project uses the ESP32-S3 as its core, paired with a custom 3D printed case and a magnetic stand to create a high-quality smart knob controller.



1. Prepare ESP32-S3 Touch Screen



2. Install 3D Printed Case & Cable



3. Attach to Magnetic Phone Stand

⚙️ Software Operation

- Value Adjustment** When you drag any curved area or touch the "Heating" or "Fan" label, the knob will immediately lock onto that curved area. You can then adjust the values using the **physical knob** or **touch controls**.
- Screen Switch** Touching the central **temperature tab** switches between screens 1 and 2.
- Unit Toggle** **Long press** the temperature tab to switch the temperature unit (°C/°F).
- Screen Reset** **Long press** the physical button to reset the screen; this method can be used to reposition the UI when it shifts.

- **Cooling Auto Shutdown**

When the cooling fan (COOL) is activated, if the temperature drops from a high setting to **below 50°C (122°F)**, the system will automatically execute the `shutdown()` function, turning off all heaters, fans, and motors. This function only works in cooling mode to avoid affecting the normal baking process.



Parts & Downloads

ESP32-S3 Board & Touchscreen	Amazon Link
Cell Phone Holder (Base)	Amazon Link
3D Print File for ESP32 Case	MakerWorld Link